SWAPD453 - Spring 2026
IoT

– SmartEnv –

Assignment Report Report

Github Repo Link

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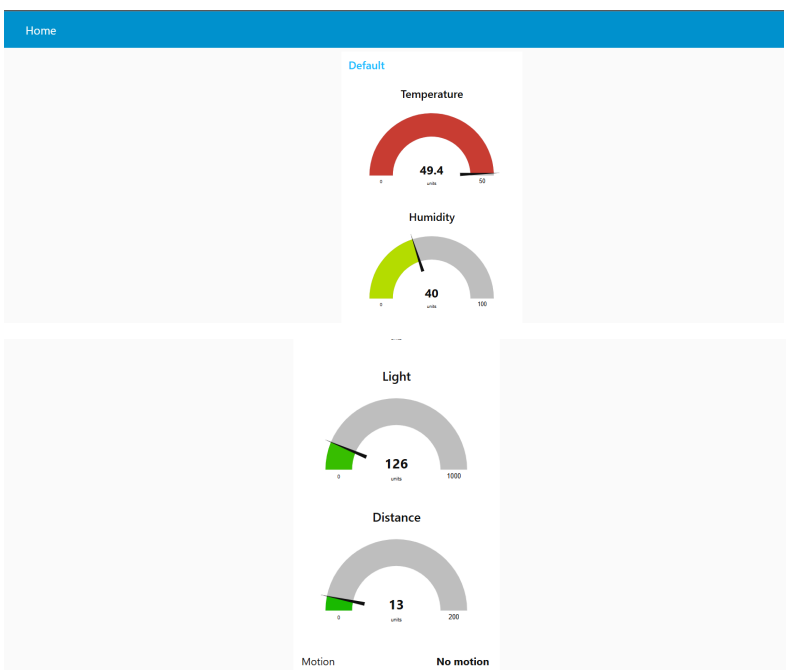
16/03/2026

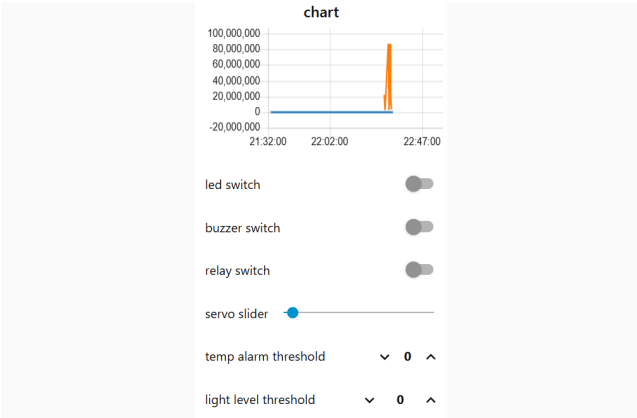
System Description

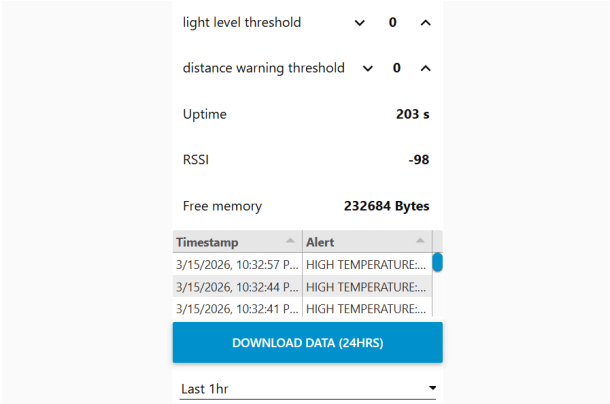
This report describes a Smart Environment Monitoring & Control System that is built in 3 layers:

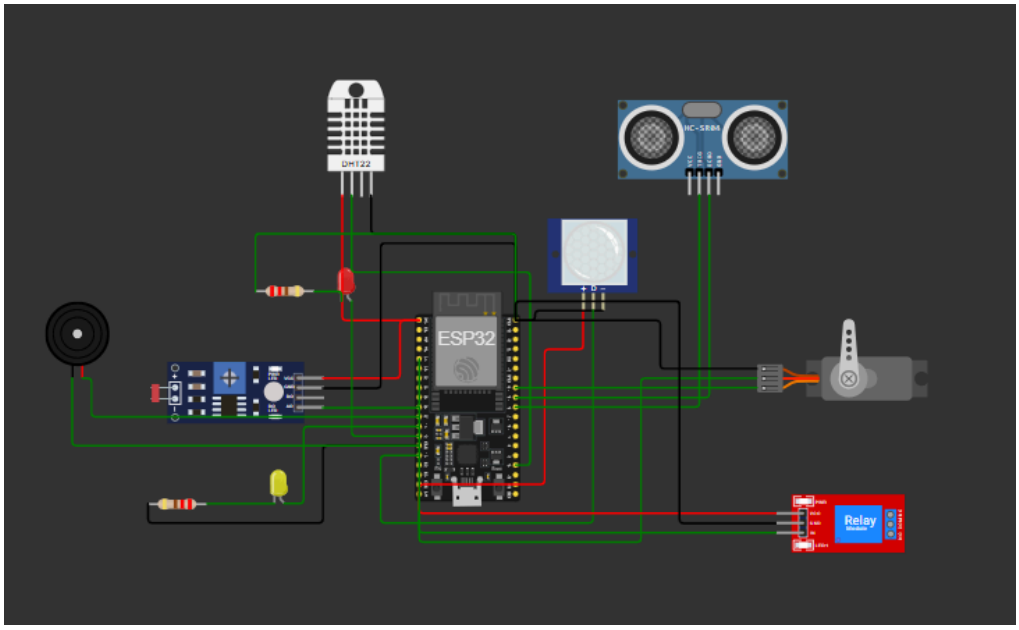
1. **Sensing layer**
   Using ESP32 microcontroller and 4 sensors: DHT22 (temperature and humidity), LDR (light), PIR(motion detection), and HC-SR04 ultrasonic sensor (distance)
2. **Communication layer**
   The ESP32 connects to a local WiFI network and uses MQTT protcol to communicate with a public broker (broker.hivemq.com), where it publishes sensor data and subscribes to actuator and configuration topics
3. **Application layer**
   Node-RED is used as a server to handle the visualization of the dashboard. The dashboard includes real-time gauges, historical data charts, manual override switches for actuators (LED, Buzzer, Servo motor, relay) and sliders to dynamically adjust system thresholds. Also, it includes an automated alert system that logs critical events to a history table and monitors ESP32 connection status via a watchdog timer

System Overview









Challenges Faced

- **MQTT Payload Formatting**: ESP was publishing sensor values as raw strings before which caused parsing failures in Node-RED. So, we updated the code to serialize the data into JSON
- **Dashboard and layers overall integration:** combining multiple layers and handling multiple dashboard inputs was a challenge that we overcame by carefully managing msg.topic strings to build a single configuration payload when necessary. Also, we aligned the inputs and outputs needed by each system layer to finally integrate right with each other.
- **Alert history table:** at first, alert history table wasn't showing the data correctly. So, we formatted the table to first transform individual alert messages into an array of objects (using a separate function) to maintain a running log of events

Lessons learned

- We learned how to integrate different layers together in order to produce a final fully-functioning system
- We learned the importance of using JSON as a standard communication protocol which ensures that both the publishing side and subscribing side agree on the exact key names and data types
- We also learned the importance of system resilience by implementing a 30-second watchdog timer and system status monitors (uptime, rssi, memory)